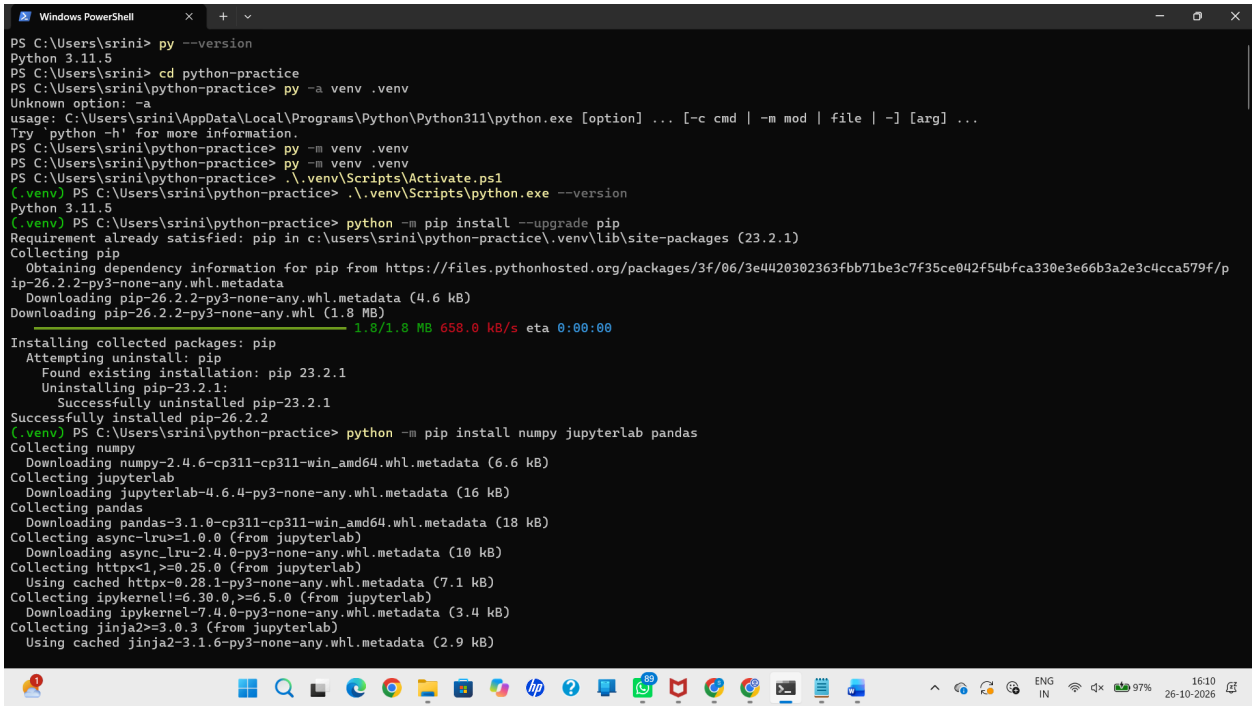
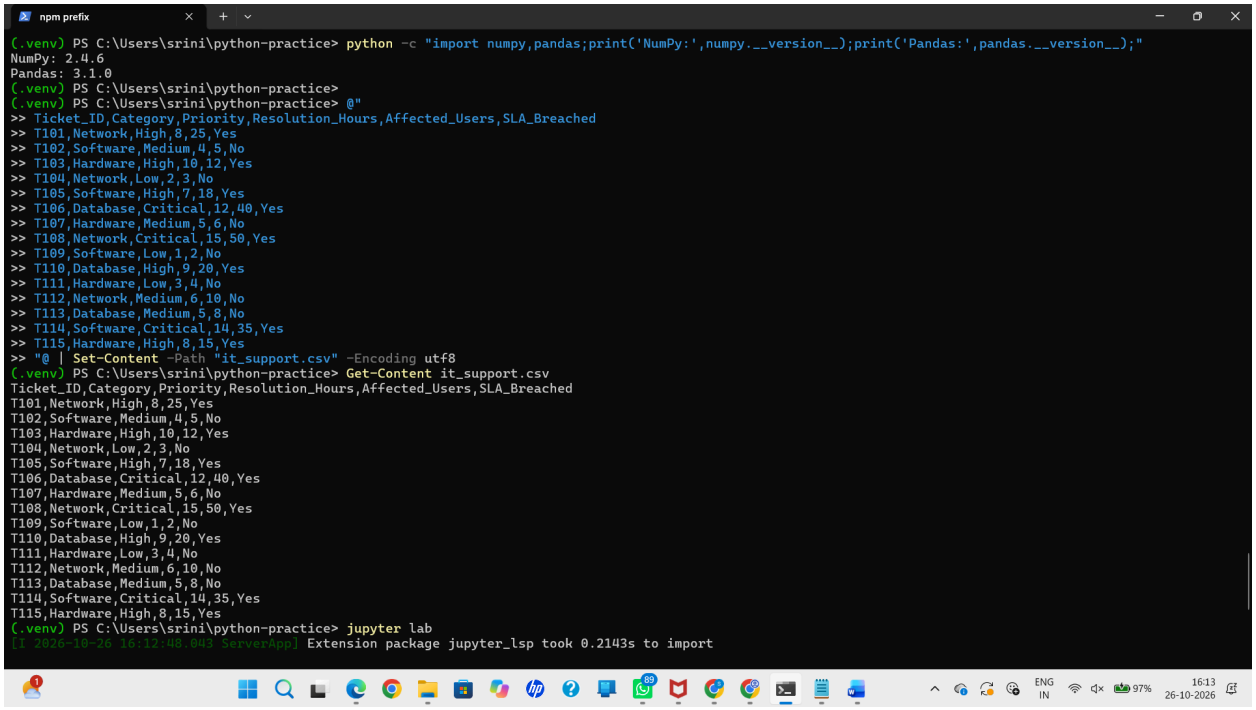
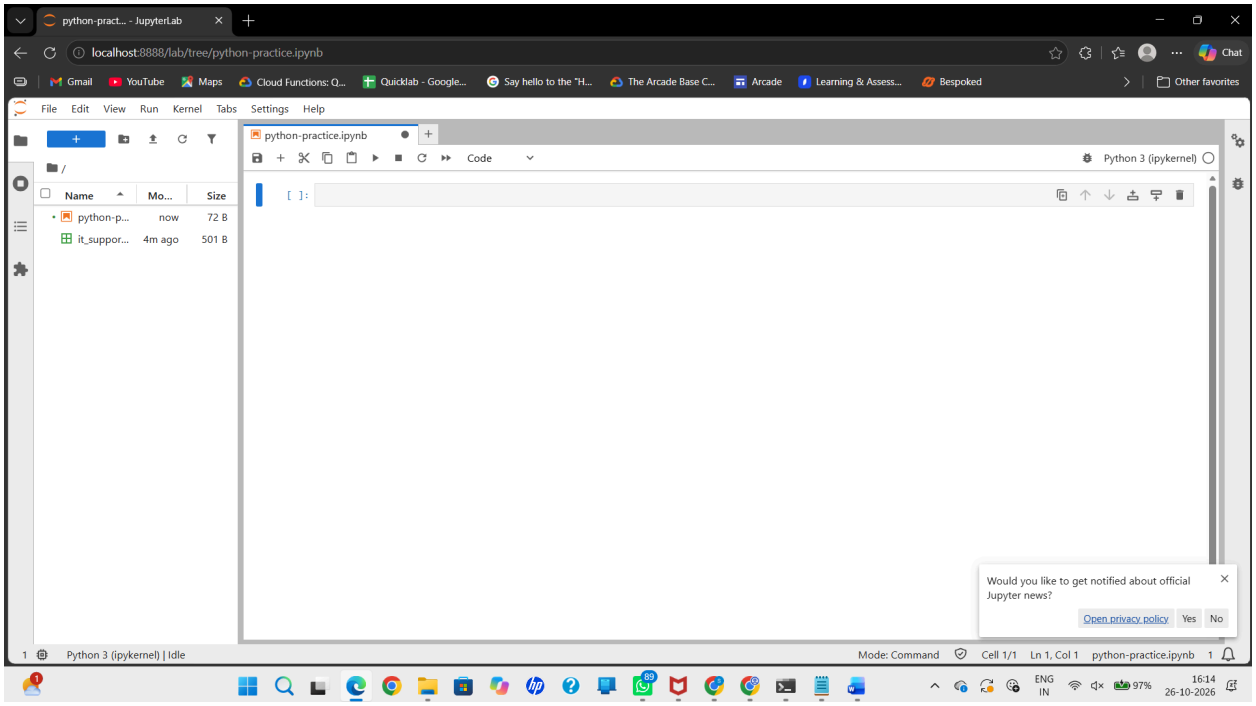
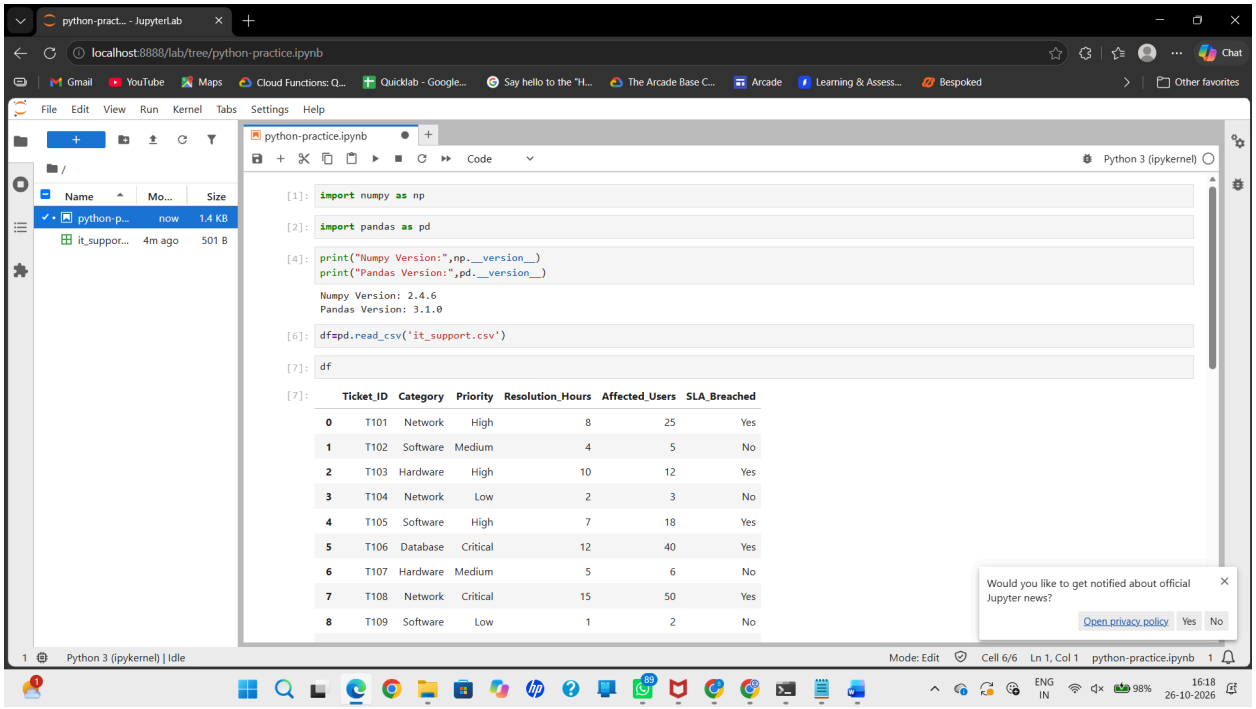
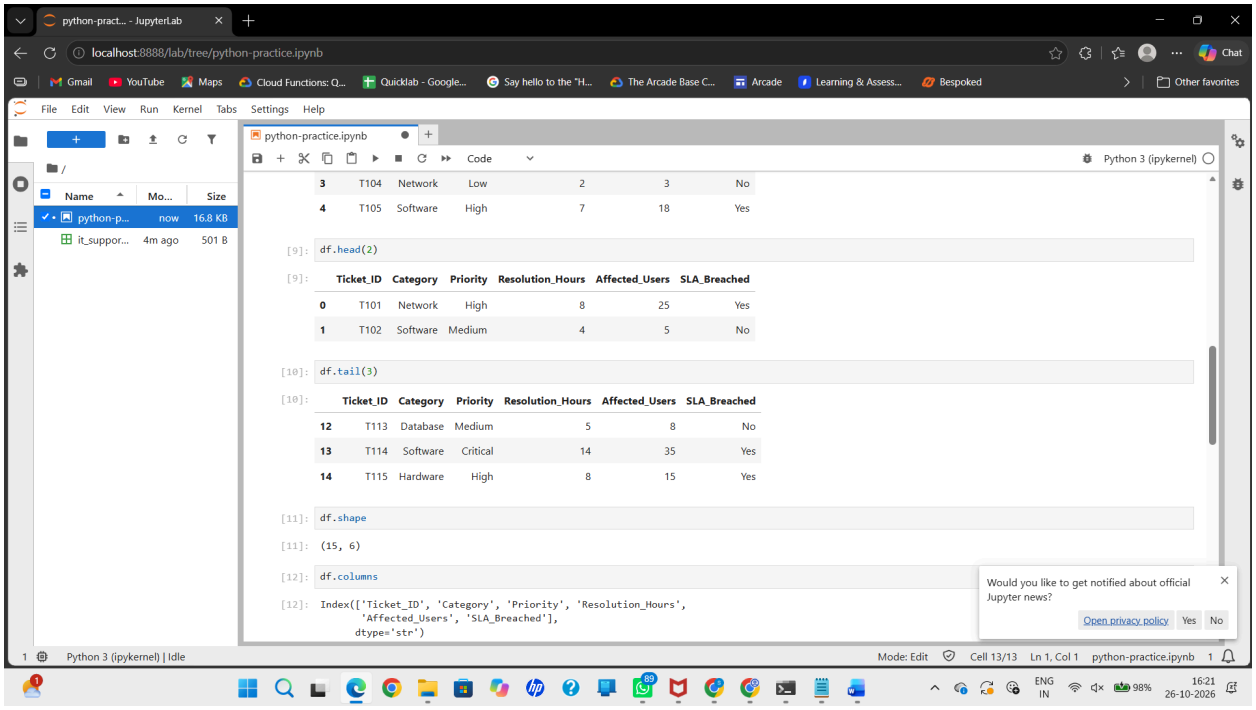
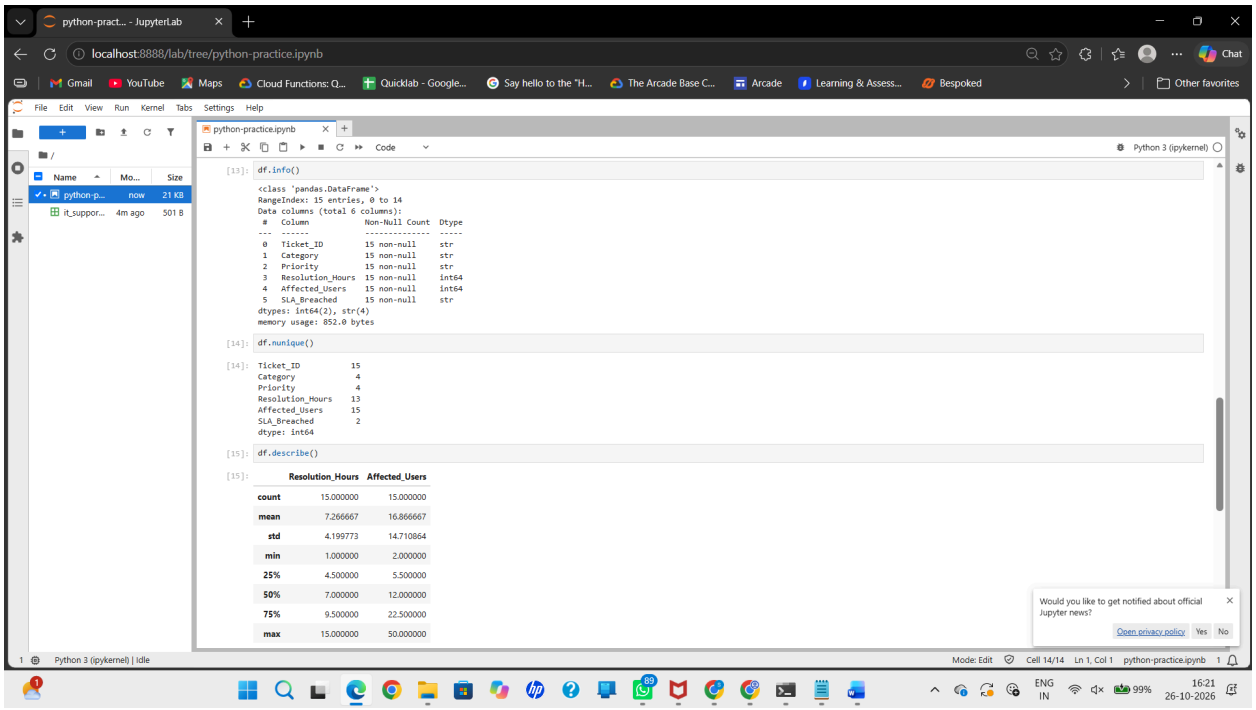

Windows PowerShell
PS C:\Users\srini> py --version
Python 3.11.5
PS C:\Users\srini> cd python-practice
PS C:\Users\srini\python-practice> py -a venv .venv
Unknown option: -a
usage: C:\Users\srini\AppData\Local\Programs\Python\Python311\python.exe [option] ... [-c cmd | -m mod | file | -] [arg] ...
Try `python -h' for more information.
PS C:\Users\srini\python-practice> py -m venv .venv
PS C:\Users\srini\python-practice> py -m venv .venv
PS C:\Users\srini\python-practice> .\.venv\Scripts\Activate.ps1
(.venv) PS C:\Users\srini\python-practice> .\.venv\Scripts\python.exe --version
Python 3.11.5
(.venv) PS C:\Users\srini\python-practice> python -m pip install --upgrade pip
Requirement already satisfied: pip in c:\users\srini\python-practice\.venv\lib\site-packages (23.2.1)
Collecting pip
Obtaining dependency information for pip from https://files.pythonhosted.org/packages/3f/06/3e44420302363fbb71be3c7f35ce042f54bfca330e3e66b3a2e3c4cca579f/pip-26.2.2-py3-none-any.whl.metadata
Downloading pip-26.2.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.2-py3-none-any.whl (1.8 MB)
1.8/1.8 MB 658.0 kB/s eta 0:00:00
Installing collected packages: pip
Attempting uninstall: pip
Found existing installation: pip 23.2.1
Uninstalling pip-23.2.1:
Successfully uninstalled pip-23.2.1
Successfully installed pip-26.2.2
(.venv) PS C:\Users\srini\python-practice> python -m pip install numpy jupyterlab pandas
Collecting numpy
Downloading numpy-2.4.6-cp311-cp311-win_amd64.whl.metadata (6.6 kB)
Collecting jupyterlab
Downloading jupyterlab-4.6.4-py3-none-any.whl.metadata (16 kB)
Collecting pandas
Downloading pandas-3.1.0-cp311-cp311-win_amd64.whl.metadata (18 kB)
Collecting async-lru>=1.0.0 (from jupyterlab)
Downloading async_lru-2.4.0-py3-none-any.whl.metadata (10 kB)
Collecting httpx<1,>=0.25.0 (from jupyterlab)
Using cached httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting ipykernel!=6.30.0,>=6.5.0 (from jupyterlab)
Downloading ipykernel-7.4.0-py3-none-any.whl.metadata (3.4 kB)
Collecting jinja2>=3.0.3 (from jupyterlab)
Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)

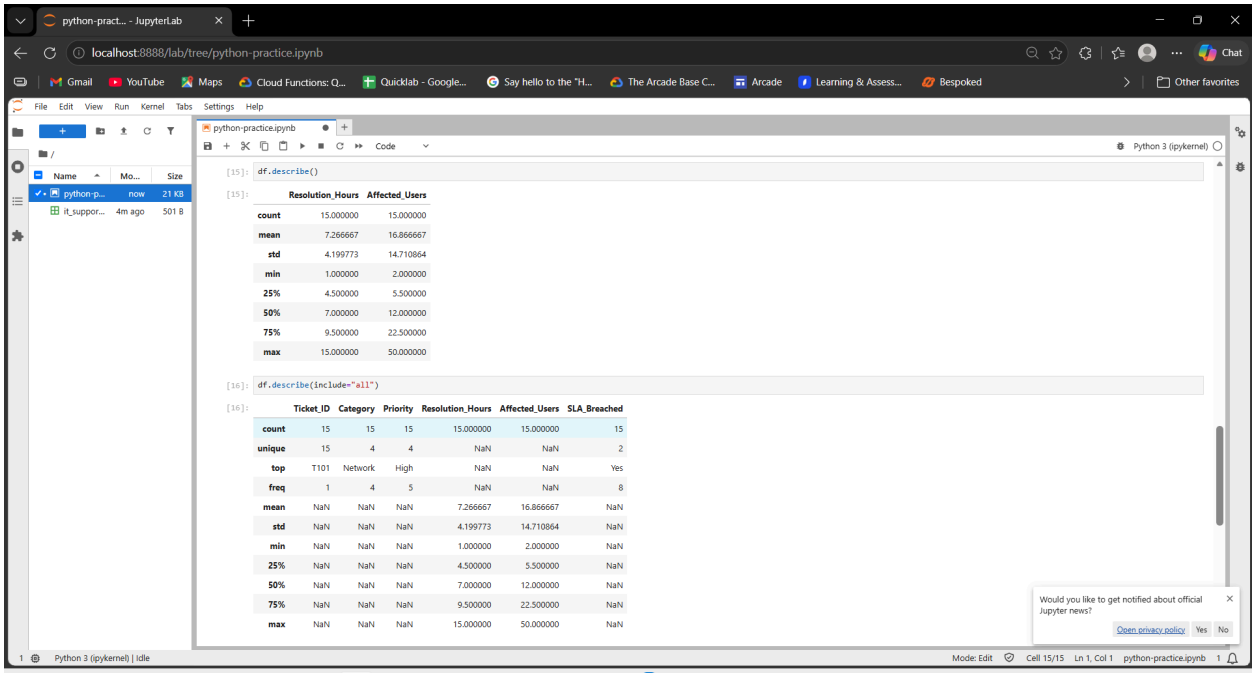

npm prefix
(.venv) PS C:\Users\srini\python-practice> python -c "import numpy,pandas;print('NumPy:',numpy.__version__);print('Pandas:',pandas.__version__);"
NumPy: 2.4.6
Pandas: 3.1.0
(.venv) PS C:\Users\srini\python-practice>
(.venv) PS C:\Users\srini\python-practice> @"
>> Ticket_ID,Category,Priority,Resolution_Hours,Affected_Users,SLA_Breached
>> T101,Network,High,8,25,Yes
>> T102,Software,Medium,4,5,No
>> T103,Hardware,High,10,12,Yes
>> T104,Network,Low,2,3,No
>> T105,Software,High,7,18,Yes
>> T106,Database,Critical,12,40,Yes
>> T107,Hardware,Medium,5,6,No
>> T108,Network,Critical,15,50,Yes
>> T109,Software,Low,1,2,No
>> T110,Database,High,9,20,Yes
>> T111,Hardware,Low,3,4,No
>> T112,Network,Medium,6,10,No
>> T113,Database,Medium,5,8,No
>> T114,Software,Critical,14,35,Yes
>> T115,Hardware,High,8,15,Yes
>> "@ | Set-Content -Path "it_support.csv" -Encoding utf8
(.venv) PS C:\Users\srini\python-practice> Get-Content it_support.csv
Ticket_ID,Category,Priority,Resolution_Hours,Affected_Users,SLA_Breached
T101,Network,High,8,25,Yes
T102,Software,Medium,4,5,No
T103,Hardware,High,10,12,Yes
T104,Network,Low,2,3,No
T105,Software,High,7,18,Yes
T106,Database,Critical,12,40,Yes
T107,Hardware,Medium,5,6,No
T108,Network,Critical,15,50,Yes
T109,Software,Low,1,2,No
T110,Database,High,9,20,Yes
T111,Hardware,Low,3,4,No
T112,Network,Medium,6,10,No
T113,Database,Medium,5,8,No
T114,Software,Critical,14,35,Yes
T115,Hardware,High,8,15,Yes
(.venv) PS C:\Users\srini\python-practice> jupyter lab
[I 2026-10-26 16:12:48.043 ServerApp] Extension package jupyter_lsp took 0.2143s to import


python-pract... - JupyterLab
localhost:8888/lab/tree/python-practice.ipynb
File Edit View Run Kernel Tabs Settings Help
python-practice.ipynb
Code
Python 3 (ipykernel)
[ ]:
Would you like to get notified about official Jupyter news?
Open privacy policy
Yes
No
Python 3 (ipykernel) | Idle
Mode: Command
Cell 1/1
Ln 1, Col 1
python-practice.ipynb


python-pract... - JupyterLab
localhost:8888/lab/tree/python-practice.ipynb
File Edit View Run Kernel Tabs Settings Help
python-practice.ipynb
Code
Python 3 (ipykernel)
[1]: import numpy as np
[2]: import pandas as pd
[4]: print("Numpy Version:",np.__version__)
print("Pandas Version:",pd.__version__)
Numpy Version: 2.4.6
Pandas Version: 3.1.0
[6]: df=pd.read_csv('it_support.csv')
[7]: df
[7]:
Ticket_ID Category Priority Resolution_Hours Affected_Users SLA_Breached
0 T101 Network High 8 25 Yes
1 T102 Software Medium 4 5 No
2 T103 Hardware High 10 12 Yes
3 T104 Network Low 2 3 No
4 T105 Software High 7 18 Yes
5 T106 Database Critical 12 40 Yes
6 T107 Hardware Medium 5 6 No
7 T108 Network Critical 15 50 Yes
8 T109 Software Low 1 2 No
Would you like to get notified about official Jupyter news?
Open privacy policy
Yes
No
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 6/6
Ln 1, Col 1
python-practice.ipynb


python-pract... - JupyterLab
localhost:8888/lab/tree/python-practice.ipynb
File Edit View Run Kernel Tabs Settings Help
python-practice.ipynb
Code
Python 3 (ipykernel)
3 T104 Network Low 2 3 No
4 T105 Software High 7 18 Yes
[9]: df.head(2)
[9]: Ticket_ID Category Priority Resolution_Hours Affected_Users SLA_Breached
0 T101 Network High 8 25 Yes
1 T102 Software Medium 4 5 No
[10]: df.tail(3)
[10]: Ticket_ID Category Priority Resolution_Hours Affected_Users SLA_Breached
12 T113 Database Medium 5 8 No
13 T114 Software Critical 14 35 Yes
14 T115 Hardware High 8 15 Yes
[11]: df.shape
[11]: (15, 6)
[12]: df.columns
[12]: Index(['Ticket_ID', 'Category', 'Priority', 'Resolution_Hours',
'Affected_Users', 'SLA_Breached'],
dtype='str')
Would you like to get notified about official Jupyter news?
Open privacy policy Yes No
Python 3 (ipykernel) | Idle
Mode: Edit Cell 13/13 Ln 1, Col 1 python-practice.ipynb


python-pract... - JupyterLab
localhost:8888/lab/tree/python-practice.ipynb
python-practice.ipynb
[13]: df.info()
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 6 columns):
# Column Non-Null Count Dtype
0 Ticket_ID 15 non-null str
1 Category 15 non-null str
2 Priority 15 non-null str
3 Resolution_Hours 15 non-null int64
4 Affected_Users 15 non-null int64
5 SLA_Breached 15 non-null str
dtypes: int64(2), str(4)
memory usage: 852.0 bytes
[14]: df.nunique()
[14]: Ticket_ID 15
Category 4
Priority 4
Resolution_Hours 13
Affected_Users 15
SLA_Breached 2
dtype: int64
[15]: df.describe()
[15]: Resolution_Hours Affected_Users
count 15.000000 15.000000
mean 7.266667 16.866667
std 4.199773 14.710864
min 1.000000 2.000000
25% 4.500000 5.500000
50% 7.000000 12.000000
75% 9.500000 22.500000
max 15.000000 50.000000
Would you like to get notified about official Jupyter news?
Open privacy policy Yes No
Python 3 (ipykernel) | Idle
Mode: Edit Cell 14/14 Ln 1, Col 1 python-practice.ipynb

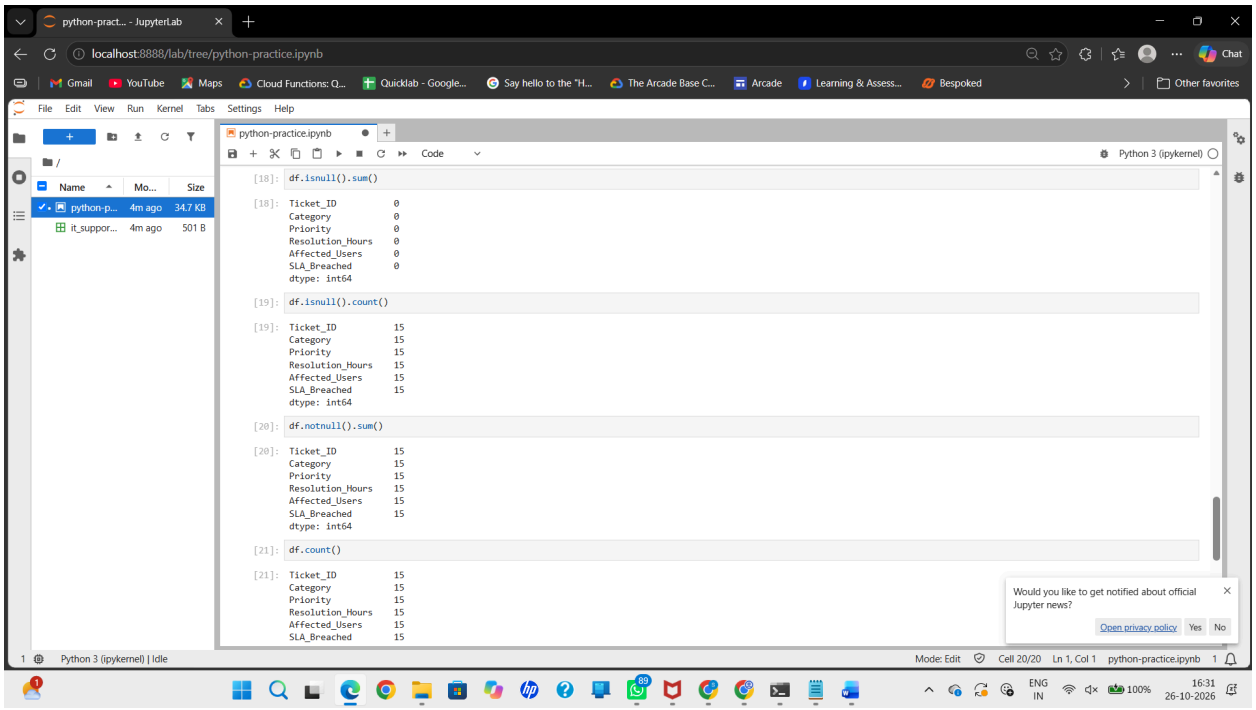


```
[15]: df.describe()
```

| | Resolution_Hours | Affected_Users |
|---|---|---|
| count | 15.000000 | 15.000000 |
| mean | 7.266667 | 16.866667 |
| std | 4.199773 | 14.710864 |
| min | 1.000000 | 2.000000 |
| 25% | 4.500000 | 5.500000 |
| 50% | 7.000000 | 12.000000 |
| 75% | 9.500000 | 22.500000 |
| max | 15.000000 | 50.000000 |

```
[16]: df.describe(include="all")
```

| | Ticket_ID | Category | Priority | Resolution_Hours | Affected_Users | SLA_Breached |
|---|---|---|---|---|---|---|
| count | 15 | 15 | 15 | 15.000000 | 15.000000 | 15 |
| unique | 15 | 4 | 4 | NaN | NaN | 2 |
| top | T101 | Network | High | NaN | NaN | Yes |
| freq | 1 | 4 | 5 | NaN | NaN | 8 |
| mean | NaN | NaN | NaN | 7.266667 | 16.866667 | NaN |
| std | NaN | NaN | NaN | 4.199773 | 14.710864 | NaN |
| min | NaN | NaN | NaN | 1.000000 | 2.000000 | NaN |
| 25% | NaN | NaN | NaN | 4.500000 | 5.500000 | NaN |
| 50% | NaN | NaN | NaN | 7.000000 | 12.000000 | NaN |
| 75% | NaN | NaN | NaN | 9.500000 | 22.500000 | NaN |
| max | NaN | NaN | NaN | 15.000000 | 50.000000 | NaN |



```
[18]: df.isnull().sum()
```

```
Ticket_ID          0
Category           0
Priority           0
Resolution_Hours   0
Affected_Users     0
SLA_Breached       0
dtype: int64
```

```
[19]: df.isnull().count()
```

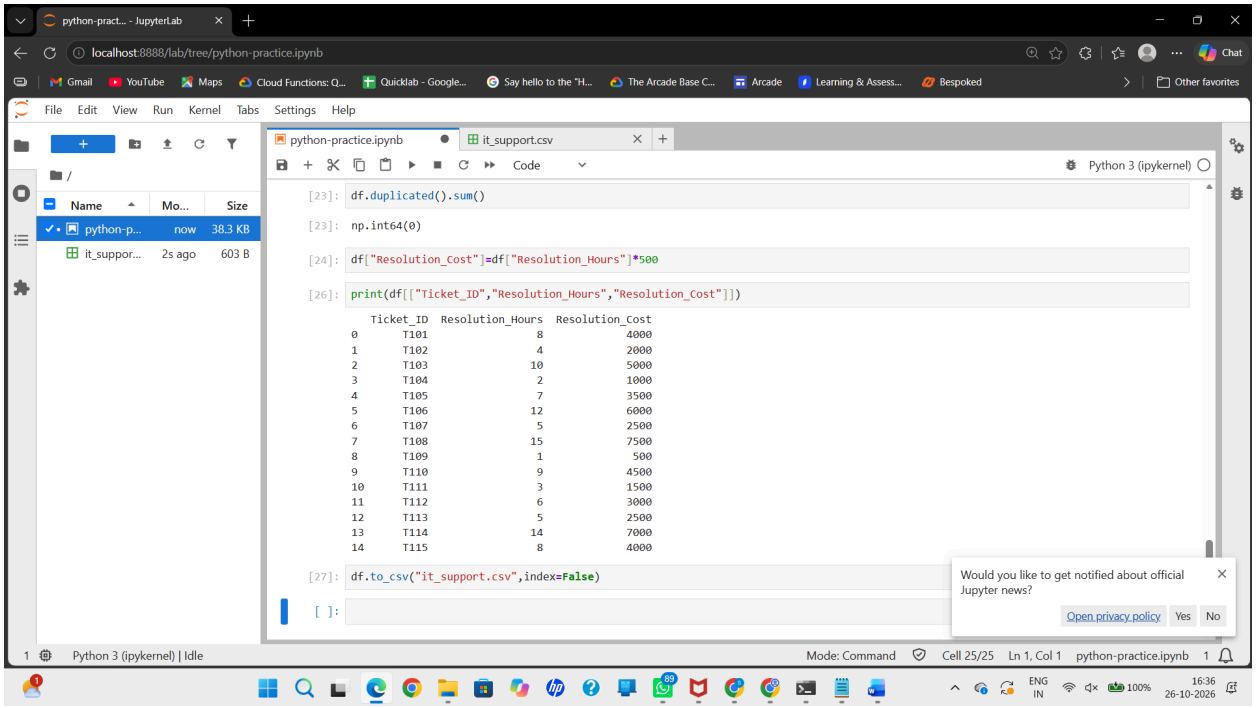
```
Ticket_ID          15
Category           15
Priority           15
Resolution_Hours   15
Affected_Users     15
SLA_Breached       15
dtype: int64
```

```
[20]: df.notnull().sum()
```

```
Ticket_ID          15
Category           15
Priority           15
Resolution_Hours   15
Affected_Users     15
SLA_Breached       15
dtype: int64
```

```
[21]: df.count()
```

```
Ticket_ID          15
Category           15
Priority           15
Resolution_Hours   15
Affected_Users     15
SLA_Breached       15
```



```
[23]: df.duplicated().sum()

[23]: np.int64(0)

[24]: df["Resolution_Cost"]=df["Resolution_Hours"]*500

[26]: print(df[["Ticket_ID","Resolution_Hours","Resolution_Cost"]])

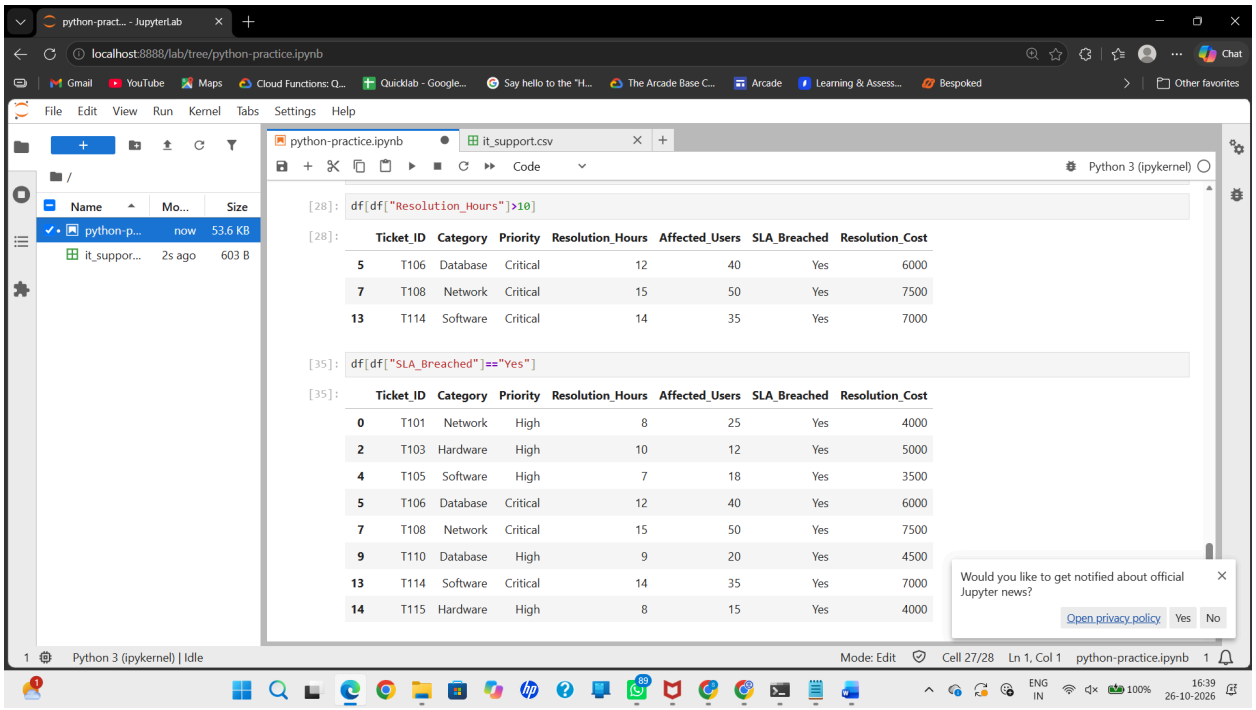
   Ticket_ID  Resolution_Hours  Resolution_Cost
0       T101                 8             4000
1       T102                 4             2000
2       T103                10             5000
3       T104                 2             1000
4       T105                 7             3500
5       T106                12             6000
6       T107                 5             2500
7       T108                15             7500
8       T109                 1              500
9       T110                 9             4500
10      T111                 3             1500
11      T112                 6             3000
12      T113                 5             2500
13      T114                14             7000
14      T115                 8             4000

[27]: df.to_csv("it_support.csv",index=False)

[ ]:
```

Would you like to get notified about official Jupyter news?

Open privacy policy | Yes | No



```
[28]: df[df["Resolution_Hours"]>10]
```

[28]:

| | Ticket_ID | Category | Priority | Resolution_Hours | Affected_Users | SLA_Breached | Resolution_Cost |
|---|---|---|---|---|---|---|---|
| 5 | T106 | Database | Critical | 12 | 40 | Yes | 6000 |
| 7 | T108 | Network | Critical | 15 | 50 | Yes | 7500 |
| 13 | T114 | Software | Critical | 14 | 35 | Yes | 7000 |

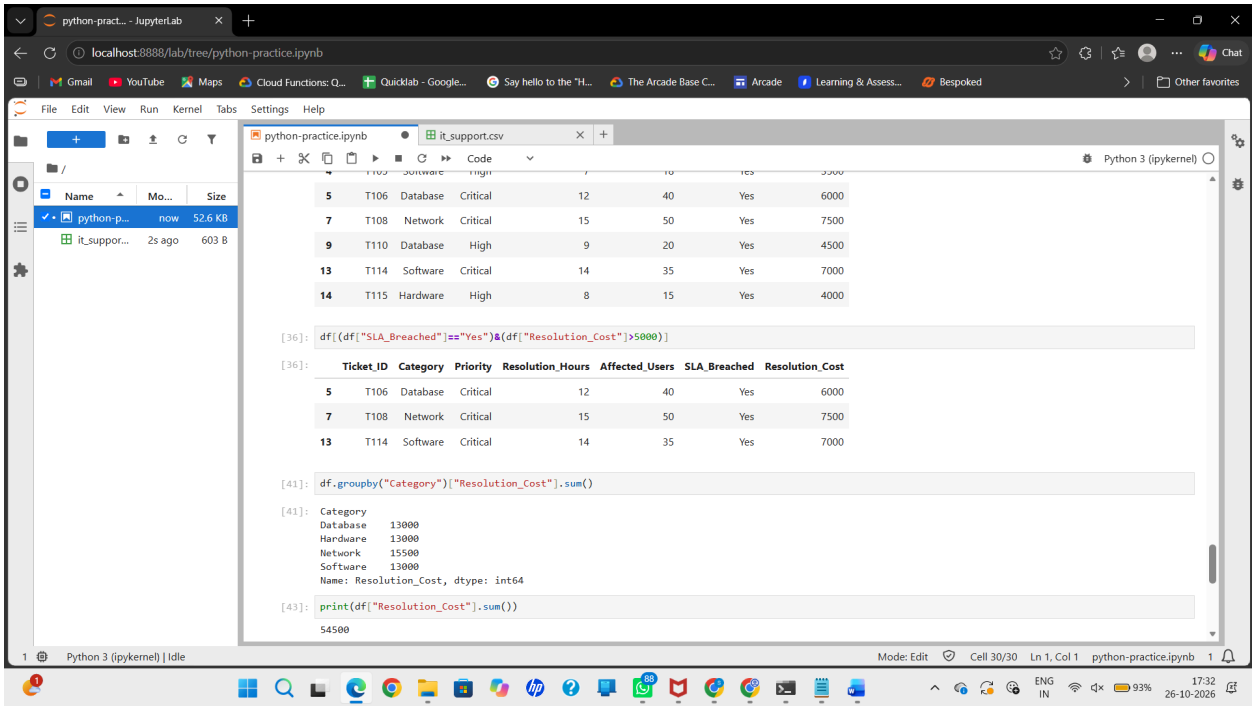
```
[35]: df[df["SLA_Breached"]=="Yes"]
```

[35]:

| | Ticket_ID | Category | Priority | Resolution_Hours | Affected_Users | SLA_Breached | Resolution_Cost |
|---|---|---|---|---|---|---|---|
| 0 | T101 | Network | High | 8 | 25 | Yes | 4000 |
| 2 | T103 | Hardware | High | 10 | 12 | Yes | 5000 |
| 4 | T105 | Software | High | 7 | 18 | Yes | 3500 |
| 5 | T106 | Database | Critical | 12 | 40 | Yes | 6000 |
| 7 | T108 | Network | Critical | 15 | 50 | Yes | 7500 |
| 9 | T110 | Database | High | 9 | 20 | Yes | 4500 |
| 13 | T114 | Software | Critical | 14 | 35 | Yes | 7000 |
| 14 | T115 | Hardware | High | 8 | 15 | Yes | 4000 |

Would you like to get notified about official Jupyter news?

Open privacy policy | Yes | No



| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 5 | T106 | Database | Critical | 12 | 40 | Yes | 6000 |
| 7 | T108 | Network | Critical | 15 | 50 | Yes | 7500 |
| 9 | T110 | Database | High | 9 | 20 | Yes | 4500 |
| 13 | T114 | Software | Critical | 14 | 35 | Yes | 7000 |
| 14 | T115 | Hardware | High | 8 | 15 | Yes | 4000 |

```
[36]: df[(df["SLA_Breached"]=="Yes")&(df["Resolution_Cost"]>5000)]
```

[36]:

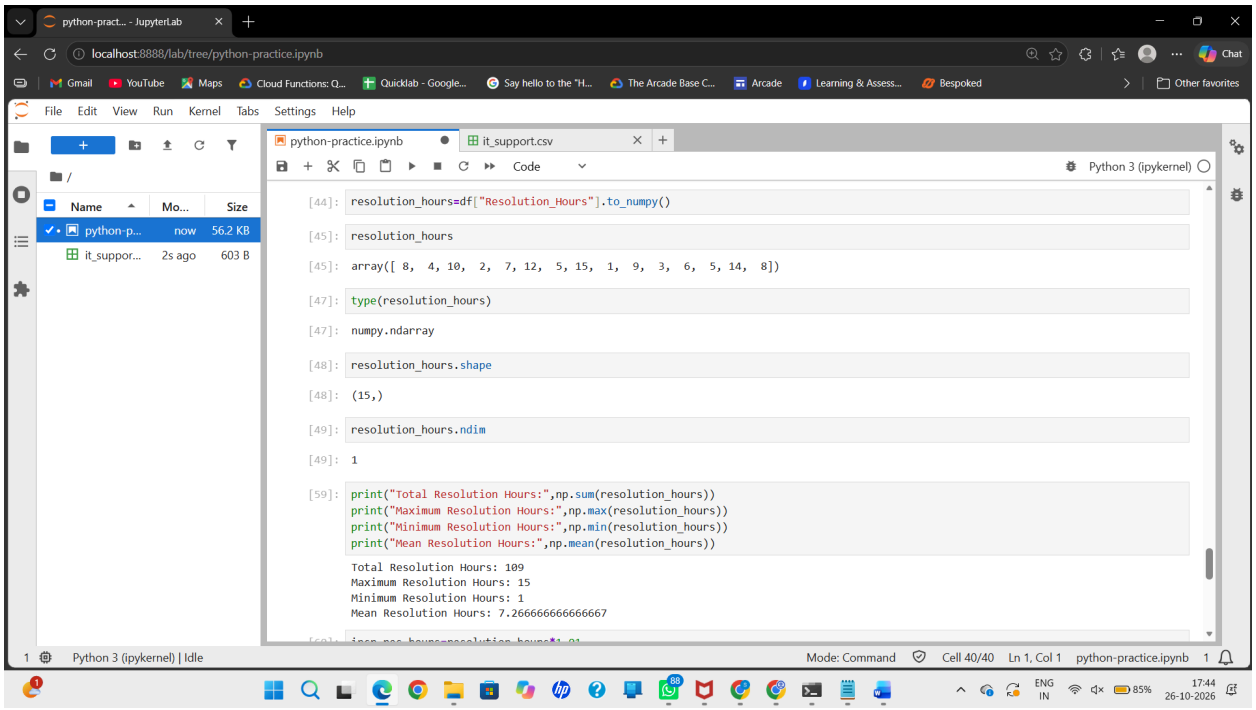
| | Ticket_ID | Category | Priority | Resolution_Hours | Affected_Users | SLA_Breached | Resolution_Cost |
|---|---|---|---|---|---|---|---|
| 5 | T106 | Database | Critical | 12 | 40 | Yes | 6000 |
| 7 | T108 | Network | Critical | 15 | 50 | Yes | 7500 |
| 13 | T114 | Software | Critical | 14 | 35 | Yes | 7000 |

```
[41]: df.groupby("Category")["Resolution_Cost"].sum()
```

```
[41]: Category
      Database    13000
      Hardware    13000
      Network     15500
      Software    13000
      Name: Resolution_Cost, dtype: int64
```

```
[43]: print(df["Resolution_Cost"].sum())
```

```
54500
```



```
[44]: resolution_hours=df["Resolution_Hours"].to_numpy()
```

```
[45]: resolution_hours
```

```
[45]: array([ 8,  4, 10,  2,  7, 12,  5, 15,  1,  9,  3,  6,  5, 14,  8])
```

```
[47]: type(resolution_hours)
```

```
[47]: numpy.ndarray
```

```
[48]: resolution_hours.shape
```

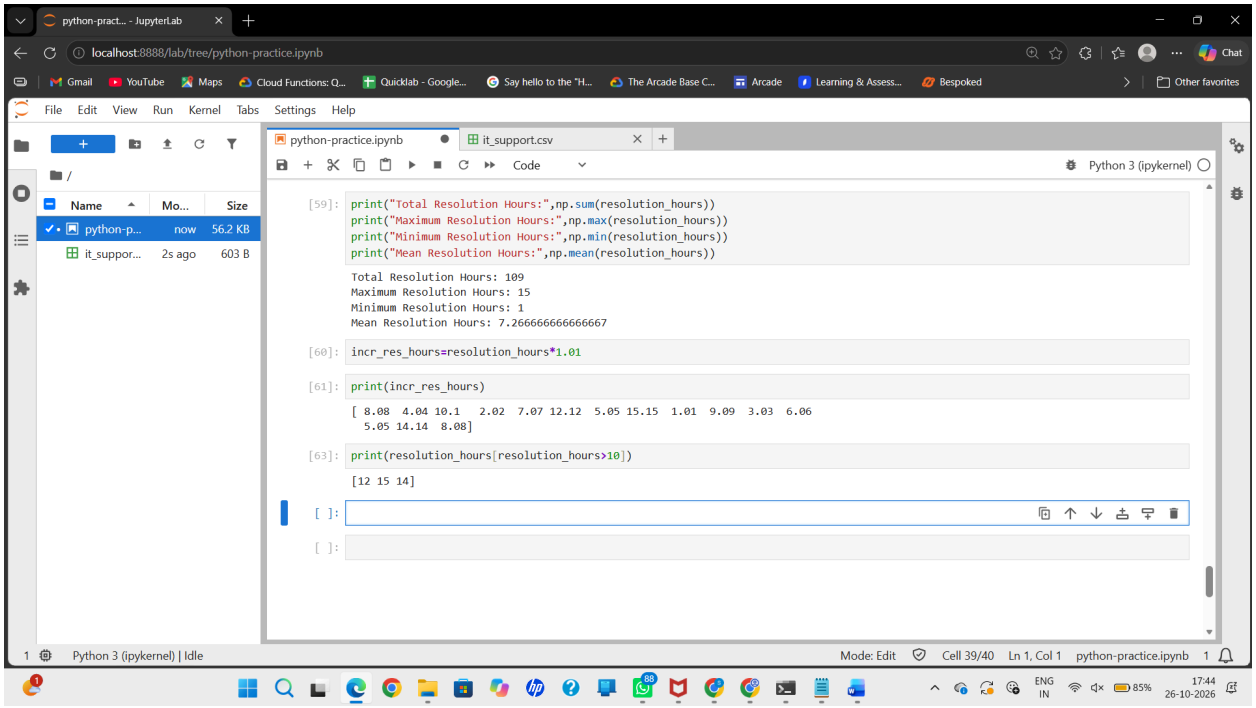
```
[48]: (15,)
```

```
[49]: resolution_hours.ndim
```

```
[49]: 1
```

```
[59]: print("Total Resolution Hours:",np.sum(resolution_hours))
      print("Maximum Resolution Hours:",np.max(resolution_hours))
      print("Minimum Resolution Hours:",np.min(resolution_hours))
      print("Mean Resolution Hours:",np.mean(resolution_hours))
```

```
Total Resolution Hours: 109
Maximum Resolution Hours: 15
Minimum Resolution Hours: 1
Mean Resolution Hours: 7.266666666666667
```


python-pract... - JupyterLab
localhost:8888/lab/tree/python-practice.ipynb
File Edit View Run Kernel Tabs Settings Help
python-practice.ipynb
it_support.csv
Code
Python 3 (ipykernel)
Name
Mo...
Size
python-p...
now
56.2 KB
it_suppor...
2s ago
603 B
[59]: print("Total Resolution Hours:",np.sum(resolution_hours))
print("Maximum Resolution Hours:",np.max(resolution_hours))
print("Minimum Resolution Hours:",np.min(resolution_hours))
print("Mean Resolution Hours:",np.mean(resolution_hours))
Total Resolution Hours: 109
Maximum Resolution Hours: 15
Minimum Resolution Hours: 1
Mean Resolution Hours: 7.266666666666667
[60]: incr_res_hours=resolution_hours*1.01
[61]: print(incr_res_hours)
[ 8.08 4.04 10.1 2.02 7.07 12.12 5.05 15.15 1.01 9.09 3.03 6.06
5.05 14.14 8.08]
[63]: print(resolution_hours[resolution_hours>10])
[12 15 14]
[ ]:
[ ]:
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 39/40
Ln 1, Col 1
python-practice.ipynb